

User-Irrepressible Sequences for Multiple-Packet Reception with MPR Capability 2: Constructions and Age of Information

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Abstract—This paper studies transmission schemes based on protocol sequences for multiple-packet reception (MPR) channels without feedback or synchronization. Unlike stochastic schemes, the proposed deterministic approach allows each user to schedule transmissions according to a preassigned sequence. A set of sequences is said to have the user-irrepressible property for MPR capability 2 (2-UI) if it provides a hard guarantee that each user achieves at least one successful transmission within a fixed period of time. We propose a new construction of 2-UI sequences for an arbitrary number of users K , which can be viewed as a generalization of the well-known prime sequences for k being a prime. By a detailed investigation of the sequence structure, we develop an efficient method to evaluate the average age of information (AoI). Simulation results demonstrate that the proposed 2-UI scheme outperforms slotted ALOHA in terms of average AoI.

I. INTRODUCTION

The rapid growth of mission-critical applications, such as the Internet of Things (IoT), demands highly reliable and timely data transmission. This paper considers the transmission network in a collision channel without feedback [1]. *Slotted ALOHA* [2] is one of the fundamental communication schemes, where each user adopts a fixed transmission probability all the time. However, the randomness of slotted ALOHA does not avoid a non-zero probability of packet loss without feedback. So, we revisit *protocol sequences* [1], which are proposed for deterministic transmission to avoid packet loss relying on the carefully designed sequence structure. This favorable property has been shown to be helpful to improve throughput and delay performance [3], [4].

Consider an uplink network where K users want to transmit their status updates (packets) to an access point (AP). The time is divided into equal time slots whose duration corresponds to the transmission time for one packet. Due to clock error and feedback-free assumption, there are relative time offsets among users. By applying protocol sequences, each user reads out the preassigned binary sequence entry sequentially and transmits a packet within a time slot if and only if the sequence value is equal to 1. A set of sequences

is said to enjoy *user-irrepressible (UI)* property [5], [6] if each sequence can permit its affiliated user to transmit at least one conflict-free packet per sequence period, regardless of the time offsets among users.

With the rapid development of the application of Internet of Things (IoT), the optimization of information freshness is getting important in real-time communication. Different from the conventional latency metric, *age of information (AoI)* has been introduced in [7] to measure the information freshness. Moreover, this paper also considers the *multiple-packet reception (MPR)* [8]–[10] transmission model that a packet can be received successfully if and only if at most $(\gamma - 1)$ other packets are being transmitted simultaneously, where γ refers to the MPR capability. This paper focuses on the UI property and the evaluation of average AoI (AAoI) for an MPR model with MPR capability $\gamma = 2$.

The evaluation of AAoI based on *CRT sequences* [6] for the traditional collision channel was proposed in [11], which focuses on periodic-status-update traffic with period equal to the length of the sequences. AAoI analysis based on *prime sequences* [12] for MPR capability 2 was given in [13], which considers the generate-at-will traffic. In addition, there also exist sequence-based schemes focusing on perfect synchronization [14], [15], but the approaches therein become inapplicable in the absence of synchronization.

In this paper, a new set of protocol sequences is designed for MPR capability 2. We evaluate the AAoI based on this newly-defined sequences both for period-status-update and generate-at-will traffics. Therefore, from the MPR perspective, this paper extends the results of [11] and can also be seen as a generalization of [13], which is restricted to prime values of K , while our results hold for arbitrary positive integers K .

II. PRELIMINARIES

A. Notation and Definitions

For any positive integer L , let $\mathbb{Z}_L \triangleq \{0, 1, \dots, L - 1\}$ denote the ring of residues modulo L . Let $\mathbf{s} = s(0)s(1)\cdots s(L - 1)$ be a binary sequence of length L .

For any non-negative integer τ , denote by $s^\tau \triangleq s(\tau)s(\tau+1)\cdots s(\tau+L-1)$ the *cyclic shift* of s by τ , where the addition is performed in $\mathbb{Z}_L$. For convenience, τ is normalized as in $\mathbb{Z}_L$. Since s can be realized as its *characteristic set* $\mathcal{I}_s \triangleq \{t \in \mathbb{Z}_L : s(t) = 1\}$, the characteristic set of s^τ can be given by $\mathcal{I}_{s^\tau} + \tau \triangleq \{t + \tau \bmod L : t \in \mathcal{I}_s\}$. Furthermore, define the *set of (non-zero) differences* of s by $d^*(s) \triangleq \{x - y \bmod L : x, y \in \mathcal{I}_s, x \neq y\}$.

For any two distinct sequences s_i and s_j , their *Hamming cross-correlation* is defined by

$$H_{s_i, s_j} \triangleq \max_{\tau \in \mathbb{Z}_L} |\mathcal{I}_{s_i} \cap (\mathcal{I}_{s_j} + \tau)|.$$

By definition, $H_{s_i, s_j} = H_{s_j, s_i}$. Obviously, $H_{s_i, s_j} = 1$ if and only if $d^*(s_i) \cap d^*(s_j) = \emptyset$.

For any set $\mathcal{A}$ and a positive integer m , let $\mathcal{A}^m$ denote the multiset containing m copies of all elements in $\mathcal{A}$. The formal definition of UI sequences under γ -MPR, called γ -UI sequences, is given below.

Definition 1. A collection $\mathcal{C} = \{C_1, \dots, C_K\}$ of subsets of $\mathbb{Z}_L$ is said to be γ -UI if

$$C_j^\gamma \not\subseteq \bigcup_{k \neq j} (C_k + \tau_k), \quad \forall j, \quad (1)$$

for any $\tau_1, \dots, \tau_K \in \mathbb{Z}_L$, where the sets are considered as multisets.

B. Evolution of the AoI

Let $A_i(t)$ denote the instantaneous AoI of user i at the beginning of slot t . Adopting the commonly used definition of $A_i(t)$ in [16], it grows linearly over time slot and drops to $t - g_i(t)$ as soon as a packet sent by user i is successfully received by the AP, where $g_i(t)$ denotes the time at which this packet was generated. As a result, we have

$$A_i(t) \triangleq \begin{cases} t - g_i(t), & \text{if user } i \text{ transmits an up-} \\ & \text{date successfully at slot } t, \\ A_i(t-1) + 1, & \text{otherwise.} \end{cases} \quad (2)$$

The average AoI (AAoI) of user i is defined by

$$\overline{A_i} \triangleq \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T A_i(t). \quad (3)$$

Let t_i^q denote the slot index when user i successfully transmits its update for the q -th time, for $q = 1, 2, \dots$. For the update successfully received at t_i^q , define its *service time* S_i^q by the time interval between its generation time and t_i^q . We further denote by Y_i^q the inter-departure time between the q -th and $(q+1)$ -th successfully transmitted updates, i.e., $Y_i^q = t_i^{q+1} - t_i^q$. By [11], one has

$$\overline{A_i} = \frac{\mathbb{E}[S_i^q Y_i^q]}{\mathbb{E}[Y_i^q]} + \frac{\mathbb{E}[(Y_i^q)^2]}{2\mathbb{E}[Y_i^q]} - \frac{1}{2}. \quad (4)$$

III. 2-UI SEQUENCES

We use $2\text{-CRT}(K)$ to denote the set of sequences constructed for K users. The construction is divided into two cases, depending on whether K is prime. The sequence length is $p_K K$, where p_K is the smallest prime number greater than or equal to K . Note that in practical applications, each of the K users is assigned a unique and distinct sequence from $2\text{-CRT}(K)$ to schedule their packet transmissions.

A. K is prime

When $K = p$ is a prime, the set $2\text{-CRT}(p)$ consists of $p+1$ sequences: $s_0, s_1, \dots, s_p$.

Definition 2. For $g = 0, 1, \dots, p-1$, let

$$\mathcal{I}_{s_g} \triangleq \{j(1+gp) \bmod p^2 : j = 0, 1, \dots, p-1\}; \quad (5)$$

and let

$$\mathcal{I}_{s_p} \triangleq \{j \cdot p \bmod p^2 : j = 0, 1, \dots, p-1\}. \quad (6)$$

The elements $(1+gp)$ in (5) and p in (6) are called the *generators* of the sequences they belong to.

From (5) and (6), for $g = 0, 1, \dots, p-1$,

$$d^*(\mathcal{I}_{s_g}) = \{\pm j(1+gp) \bmod p^2 : j = 1, \dots, p-1\}; \quad (7)$$

and

$$d^*(\mathcal{I}_{s_p}) = \{p, 2p, \dots, p^2 - p\}. \quad (8)$$

For a prime p , let $\mathbb{F}_p$ denote the finite field of order p .

Proposition 1. Let p be a prime and g, h be integers with $0 \leq g < h \leq p-1$. If $i(1+gp) \equiv \pm j(1+hp) \bmod p^2$ for some $0 \leq i, j \leq p-1$, then $i = (h-g)^{-1}$ and $j = -(h-g)^{-1}$, where the inverse is performed in $\mathbb{F}_p$.

Proof. Since $g \neq h$, it must be that case that $i(1+gp) \equiv -j(1+hp) \bmod p^2$, which implies that

$$(i+j) + p(ig+jh) = p^2. \quad (9)$$

Taking modulo p , we obtain from (9) that $i+j = p$ and

$$ig+jh = p-1 \bmod p. \quad (10)$$

By plugging $i = p-j$ into (10), we have $j(h-g) \equiv -1 \bmod p$, which implies that $j = -(h-g)^{-1}$ and $i = (h-g)^{-1}$. $\square$

Lemma 1. Consider the set $2\text{-CRT}(p)$, where p is a prime. Then, $H_{s_g, s_p} = 1$ for $0 \leq g \leq p-1$ and $H_{s_g, s_h} = 2$ for $0 \leq g < h \leq p-1$.

Proof. By (7) and (8), it is easy to see that $d^*(\mathcal{I}_{s_p}) \cap d^*(\mathcal{I}_{s_g}) = \emptyset$, which implies that $H_{s_g, s_p} = 1$ for all $g \neq p$.

Consider $0 \leq g < h \leq p-1$. As there $i = (h-g)^{-1}$ always exists in $\mathbb{F}_p$ for $0 \leq g < h \leq p-1$, by Proposition 1 and (7), we have $d^*(\mathcal{I}_{s_g}) \cap d^*(\mathcal{I}_{s_h}) \neq \emptyset$, which implies that $H_{s_g, s_h} \geq 2$. Then, it suffices to show that $H_{s_g, s_h} \leq 2$.

Suppose to the contrary that $H_{s_g, s_h} \geq 3$. Assume $a(1+gp), b(1+gp)$ and $c(1+gp)$ are three elements in $\mathcal{I}_{s_g} \cap (\mathcal{I}_{s_h} + \tau)$ for some $0 \leq a < b < c \leq p-1$ and $\tau \in \mathbb{Z}_L$. Let $i = b-a$ and $i' = c-a$. Since $a(1+gp), b(1+gp) \in \mathcal{I}_{s_g} \cap (\mathcal{I}_{s_h} + \tau)$,

we have $i(1 + gp) \equiv \pm j(1 + hp) \pmod{p^2}$ for some j . By Proposition 1, $i = (h - g)^{-1}$ in $\mathbb{F}_p$. Similarly, we have $i' = (h - g)^{-1}$. These two equations conclude that $b - a = c - a$, contradicting to $a < b < c$. This completes the proof. $\square$

Theorem 1. Any p -subset $\mathcal{C} \subseteq 2\text{-CRT}(p)$ has 2-UI property.

Proof. By Lemma 1, any sequence $\mathcal{I}_{s_g}$ has at most $2(p - 1)$ overlappings with $\cup_{s_h \in \mathcal{C} \setminus s_g} (\mathcal{I}_{s_h} + \tau_h)$ for any τ_h , counting multiplicity. Since $|\mathcal{I}_s| = p$, by the Pigeonhole Principle, there exists at least one element in $\mathcal{I}_{s_g}$ that is not overlapped by other sequences at most one time, satisfying the requirement in (1). Hence, the 2-UI property holds. $\square$

Example 1. Let $p = 5$. The characteristic sets of the six sequences in $2\text{-CRT}(5)$ given by Definition 2 are:

$$\begin{aligned} \mathcal{I}_{s_0} &= \{0, 1, 2, 3, 4\}, & \mathcal{I}_{s_1} &= \{0, 6, 12, 18, 24\}, \\ \mathcal{I}_{s_2} &= \{0, 11, 22, 8, 19\}, & \mathcal{I}_{s_3} &= \{0, 16, 7, 23, 14\}, \\ \mathcal{I}_{s_4} &= \{0, 21, 17, 13, 9\}, & \mathcal{I}_{s_5} &= \{0, 5, 10, 15, 20\}. \end{aligned}$$

Remark 1. The sequences in $2\text{-CRT}(p)$ are essentially prime sequences [12], in which the difference lies only in the indexing of the sequences. In other words, we provide another proof to the 2-UI property of prime sequences.

B. K is composite

When K is composite, the set $2\text{-CRT}(K)$ consists of $K + 1$ sequences: $s_0, s_1, \dots, s_{p_K}$.

Definition 3. For $g = 0, 1, \dots, p_K - 1$, let

$$\widehat{\mathcal{I}}_{s_g} \triangleq \{j(g, 1) \in \mathbb{Z}_{p_K} \times \mathbb{Z}_K : j = 0, 1, \dots, K - 1\}; \quad (11)$$

and let

$$\widehat{\mathcal{I}}_{s_{p_K}} \triangleq \{j(1, 0) \in \mathbb{Z}_{p_K} \times \mathbb{Z}_K : j = 0, 1, \dots, K - 1\}. \quad (12)$$

Then, for $g = 0, 1, \dots, p_K$, let

$$\mathcal{I}_{s_g} = \{t \in \mathbb{Z}_{p_K K} : (t \bmod p_K, t \bmod K) \in \widehat{\mathcal{I}}_{s_g}\}. \quad (13)$$

The elements $(g, 1)$ in (11) and $(1, 0)$ in (12) are the generators.

Example 2. When $K = 4$, one has $p_K = 5$. The characteristic sets of the six sequences sets in $2\text{-CRT}(5)$ given by Definition 3 are:

$$\begin{aligned} \mathcal{I}_{s_0} &= \{0, 5, 10, 15\}, & \mathcal{I}_{s_1} &= \{0, 1, 2, 3\}, \\ \mathcal{I}_{s_2} &= \{0, 17, 14, 11\}, & \mathcal{I}_{s_3} &= \{0, 13, 6, 19\}, \\ \mathcal{I}_{s_4} &= \{0, 9, 18, 7\}, & \mathcal{I}_{s_5} &= \{0, 16, 12, 8\}. \end{aligned}$$

Note that the definition in (13) is well-defined since $\mathbb{Z}_{p_K K} \cong \mathbb{Z}_{p_K} \times \mathbb{Z}_K$ due to $\gcd(p_K, K) = 1$. More precisely, the mapping $\theta : \mathbb{Z}_{p_K K} \rightarrow \mathbb{Z}_{p_K} \times \mathbb{Z}_K$ by

$$\theta(t) = (t \bmod p_K, t \bmod K) \quad (14)$$

is a bijection. This correspondence is known as the *Chinese Remainder Theorem*. In this fashion, the cyclic shift of $\mathcal{I}_{s_g} + \tau$ can be performed by

$$\widehat{\mathcal{I}}_{s_g} + (\mu, \nu),$$

where $\tau \equiv \mu \bmod p_K$ and $\tau \equiv \nu \bmod K$. So, we also call $\widehat{\mathcal{I}}_{s_g}$ as the characteristic set of s_g with generator $\theta^{-1}(\mathbf{g})$, where $\mathbf{g}$ is the generator of $\widehat{\mathcal{I}}_{s_g}$.

From (11) and (12), for $g = 0, 1, \dots, p_K - 1$,

$$d^*(\widehat{\mathcal{I}}_{s_g}) = \{\pm j(g, 1) \in \mathbb{Z}_{p_K} \times \mathbb{Z}_K : 1 \leq j \leq K - 1\}; \quad (15)$$

and

$$d^*(\widehat{\mathcal{I}}_{s_{p_K}}) = \{\pm j(1, 0) \in \mathbb{Z}_{p_K} \times \mathbb{Z}_K : 1 \leq j \leq K - 1\}. \quad (16)$$

Proposition 2. Let K be a composite and g, h be integers with $1 \leq g < h \leq p_K - 1$. If $i(g, 1) \equiv \pm j(h, 1)$ in $\mathbb{Z}_{p_K} \times \mathbb{Z}_K$ for some $0 \leq i, j \leq p_K - 1$, then $i = K + Kg(h - g)^{-1}$ and $j = -Kg(h - g)^{-1}$, where the inverse is performed in the finite field $\mathbb{F}_p$.

Proof. Since $g \neq h$, it must be the case that

$$i(g, 1) \equiv -j(h, 1) \text{ in } \mathbb{Z}_{p_K} \times \mathbb{Z}_K,$$

which implies that

$$i + j = K \text{ and } ig + jh \equiv 0 \pmod{p}.$$

Plugging $i = K - j$ into the second equation yields that

$$(K - j)g + jh \equiv 0 \pmod{p}. \quad (17)$$

It follows that $j = -Kg(h - g)^{-1}$ and then $i = K + Kg(h - g)^{-1}$ due to $i + j = K$. $\square$

Lemma 2. Consider the set $2\text{-CRT}(K)$, where K is a composite. Then,

- (i) $H_{s_0, s_g} = H_{s_g, s_{p_K}} = 1$ for $1 \leq g \leq p_K - 1$.
- (ii) $H_{s_g, s_h} \leq 2$ for $1 \leq g < h \leq p_K - 1$, and $H_{s_g, s_h} = 2$ if and only if

$$h \equiv g - Kgj^{-1} \quad (18)$$

for some $j \in \{1, 2, \dots, K - 1\}$, where the inverse is performed in $\mathbb{F}_{p_K}$.

Proof. We only consider (ii) as (i) is obvious from (15)–(16).

Similar to the proof of Lemma 1, it can be deduced from Proposition 2 that $H_{s_g, s_h} \leq 2$ for $1 \leq g < h \leq p_K - 1$. If $H_{s_g, s_h} = 2$, by Proposition 2 again, the equation (17) holds for some $0 \leq j \leq K - 1$, which implies (18). $\square$

By analogy with Theorem 1, we have the following result.

Theorem 2. Any K -subset $\mathcal{C} \subseteq 2\text{-CRT}(K)$ has 2-UI property.

IV. AOI ANALYSIS FOR 2-CRT SEQUENCES

In this section, we further investigate the structure of 2-CRT sequences and derive explicit AoI formulas for two specific transmission traffic scenarios.

By Lemmas 1 and 2, for any K , prime or composite, the sequences in $2\text{-CRT}(K)$ can be classified into two types:

- *Type I:* those whose Hamming cross-correlation with all other sequences equals 1; and
- *Type II:* those whose Hamming cross-correlation with all other sequences is at most 2.

We study Type II sequences in more detail, as Type I sequences have a simpler structure.

Given a subset $\mathcal{A} \subseteq \mathbb{Z}_L$ and an element $x \in \mathbb{Z}_L$, define

$$f(\mathcal{A}; x) \triangleq |\{(a, b) \in \mathcal{A} \times \mathcal{A} : a - b \equiv x \pmod{L}\}|. \quad (19)$$

For two subsets $\mathcal{A}, \mathcal{B} \subseteq \mathbb{Z}_L$ such that $d^*(\mathcal{A}) \cap d^*(\mathcal{B}) = \{\pm x\}$, we use

$$e(\mathcal{A}, \mathcal{B}) \triangleq f(\mathcal{A}; x)f(\mathcal{B}; x) \quad (20)$$

to denote the number of cyclic shifts τ such that $|\mathcal{A} \cap (\mathcal{B} + \tau)| = 2$. This value plays an important role in evaluating the AoI of a user whose preassigned sequence is of Hamming cross-correlation 2 with other users.

Lemma 3. Assume $d^*(\mathcal{I}_{s_g}) \cap d^*(\mathcal{I}_{s_h}) = \{\pm x\}$. If $\{\pm x\} = \{\pm i\mathbf{g}\}$ for some $1 \leq i \leq K-1$, then

$$e(\mathcal{I}_{s_g}, \mathcal{I}_{s_h}) = i(K-i). \quad (21)$$

Moreover, for $\mathcal{A} \subset \mathcal{I}_{s_g}$

$$e(\mathcal{A}, \mathcal{I}_{s_h}) = \begin{cases} 0 & \text{if } \{\pm x\} \not\subseteq d^*(\mathcal{A}) \\ f(\mathcal{A}; x) \cdot i & \text{otherwise.} \end{cases} \quad (22)$$

Proof. Let $\mathbf{g}$ and $\mathbf{h}$ be the generators of $\mathcal{I}_{s_g}$ and $\mathcal{I}_{s_h}$, respectively. When $\{\pm x\} = \{\pm i\mathbf{g}\}$, the pairs of elements in $\mathcal{I}_{s_g}$ whose difference is $\pm x$ are $\{0, i\mathbf{g}\}, \{\mathbf{g}, (i+1)\mathbf{g}\}, \dots, \{(K-i-1)\mathbf{g}, (K-1)\mathbf{g}\}$. So, $f(\mathcal{I}_{s_g}; x) = K-i$. By the proofs of Propositions 1 and 2, those pairs of elements in $\mathcal{I}_{s_h}$ whose difference is $\pm x$ are $\{0, (K-i)\mathbf{h}\}, \dots, \{(i-1)\mathbf{h}, (K-i)\mathbf{h}\}$. So we have $f(\mathcal{I}_{s_h}; x) = i$. This proves the identity (21).

Let $\mathcal{A} \subset \mathcal{I}_{s_g}$. Since $\{\pm x\} \not\subseteq d^*(\mathcal{A})$ means that $f(\mathcal{A}; x) = 0$, then (22) follows. $\square$

We consider $T = 1$ as the fundamental generate-at-will benchmark to evaluate the impact of sequence structure on packet inter-arrival times. The case of $T = L$ is chosen because it aligns with the 2-UI property, providing a deterministic guarantee that every periodic update is successfully delivered within a single sequence period.

A. Generate-at-will traffic

In this scenario, we have $S_i^q = 0$ for all i and q . Plugging it into (4) yields

$$\overline{A_i} = \frac{\mathbb{E}[(Y_i^q)^2]}{2\mathbb{E}[Y_i^q]} - \frac{1}{2}. \quad (23)$$

Let $\mathcal{C}$ be a K -subset of 2-CRT(K). In a generate-at-will traffic model, $\mathbb{E}[Y_i^q]$ is the sequence length L divided by its throughput, denoted by $\mathcal{T}_i$. That is,

$$\mathbb{E}[Y_i^q] = L/\mathcal{T}_i. \quad (24)$$

Counting $\mathbb{E}[(Y_i^q)^2]$ is rather challenging. We therefore adopt the approach proposed in [13], which evaluates $\mathbb{E}[(Y_i^q)^2]$ from its definition. Relying on the special structure of 2-CRT sequences, the computational complexity is significantly reduced.

Fix a cyclic shift $\tau \in \mathbb{Z}_L$ and assume $\mathcal{I}_{s_i}^\tau = \{x_1, x_2, \dots, x_K\}$ with $x_1 < x_2 < \dots < x_K$. For $\mathcal{A} \subseteq \mathcal{I}_{s_i}^\tau$ and any $j \in \mathcal{C} \setminus \{i\}$, define

$$\alpha_j(\mathcal{A}) \triangleq |\{\tau' \in \mathbb{Z}_L : \mathcal{I}_{s_i}^\tau \cap \mathcal{I}_{s_j}^{\tau'} = \mathcal{A}\}|$$

as the *reverse Hamming cross-correlation*. As $H_{s_i, s_j} \leq 2$, by the same argument of the proof of [13, Lemma 2],

$$\alpha_j(\mathcal{A}) = \begin{cases} L - K^2 + e(\mathcal{I}_{s_i}, \mathcal{I}_{s_j}), & \text{if } \mathcal{A} = \emptyset, \\ K - \sum_{x' \in \mathcal{I}_{s_i} \setminus \{x\}} f(\mathcal{I}_{s_j}; x - x'), & \text{if } \mathcal{A} = \{x\}, \\ f(\mathcal{I}_{s_j}; x_a - x_b), & \text{if } \mathcal{A} = \{x_a, x_b\}. \end{cases}$$

Consider an event that the set of collided packets by user j is $\mathcal{A}_j$ for each $j \neq i$. Such an event occurs with probability

$$\prod_{j \in \mathcal{C} \setminus \{i\}} \alpha_j(\mathcal{A}_j) / L^{K-1}. \quad (25)$$

Let $\mathcal{B} = \cup_{j \in \mathcal{C} \setminus \{i\}} \mathcal{A}_j$. Given a set $\mathcal{I} = \{a_1, a_2, \dots, a_w\}$ with $0 \leq a_1 < a_2 < \dots < a_w < L$, define

$$\eta(\mathcal{I}) \triangleq (a_2 - a_1)^2 + (a_3 - a_2)^2 + \dots + (a_w - a_{w-1})^2.$$

As the set of successful packets in $\mathcal{I}_{s_i}$ under $\mathcal{B}$ is $\mathcal{I}_{s_i}^\tau \cap ((\mathcal{I}_{s_i})^2 \setminus \mathcal{B})$, by (24), the value $\mathbb{E}[(Y_i^q)^2]$ equals to

$$\frac{1}{L} \sum_{\tau=0}^L \sum_{(\mathcal{I}_{s_i})^2 \setminus \mathcal{B} \neq \emptyset} \eta(\mathcal{I}_{s_i}^\tau \cap ((\mathcal{I}_{s_i})^2 \setminus \mathcal{B})) \prod_{j \in \mathcal{C} \setminus \{i\}} \alpha_j(\mathcal{A}_j) / L^{K-1} \quad (26)$$

over the throughput of user i , $\mathcal{T}_i$. By (23) and (24), $\overline{A_i}$ is equal to

$$\frac{1}{2L^{K+1}} \sum_{\tau=0}^L \sum_{(\mathcal{I}_{s_i})^2 \setminus \mathcal{B} \neq \emptyset} \eta(\mathcal{I}_{s_i}^\tau \cap ((\mathcal{I}_{s_i})^2 \setminus \mathcal{B})) \prod_{j \in \mathcal{C} \setminus \{i\}} \alpha_j(\mathcal{A}_j) - \frac{1}{2}.$$

B. Periodic traffic

In this scenario, we only consider the period of status updates $T = L = p_K K$. Therefore, we have $\mathbb{E}[Y_i^q] = L$ and $\mathbb{E}[(Y_i^q)^2] = L^2$ for any i and q . By plugging them into (4) yields that

$$\overline{A_i} = \mathbb{E}[S_i^q] + \frac{L}{2} - \frac{1}{2}. \quad (27)$$

Note that $\mathbb{E}[S_i^q]$ is equal to the average access delay, defined by the number of slots that user i has to wait until one packet can be transmitted successfully [4].

Let $\mathcal{C}$ be a K -subset of 2-CRT(K). We first consider a Type I sequence $\mathbf{s}_g \in \mathcal{C}$, i.e., $H_{s_g, s_h} = 1$ for all $h \neq g$. Consider $\mathcal{I}_{s_g} = \{a_0, a_1, \dots, a_{K-1}\}$ a subset in $\mathbb{Z}_L$. Let $\delta = (\delta_0, \delta_1, \dots, \delta_{K-1})$ be the vector of consecutive differences of $\mathcal{I}_{s_g}$ by defining $\delta_i = a_i - a_{i-1} \pmod{L}$. For $d = 0, 1, \dots, K-1$, let

$$C_\delta(d) = \sum_{j=0}^{K-1} \delta_j \delta_{j-d} \quad (28)$$

Following [17, Theorem 1], we have

$$\mathbb{E}[S_i^q] = \frac{1}{L} \left(\frac{1}{2} C_\delta(0) + \sum_{n=1}^{K-1} (1 - h_n) C_\delta(n) \right) - \frac{1}{2}, \quad (29)$$

where

$$h_n \triangleq \sum_{i=1}^n (-1)^{i+1} \binom{n}{i} \left(1 - \frac{i}{p_K} \right)^{K-1} \quad (30)$$

is defined for $n = 1, 2, \dots, K-1$.

Let $\mathbf{s}_g \in \mathcal{C}$ be a Type II sequence. Fix a cyclic shift $\tau \in \mathbb{Z}_L$ and assume $\mathcal{I}_{\mathbf{s}_g}^\tau = \{x_1, x_2, \dots, x_K\}$ with $x_1 < x_2 < \dots < x_K$. Following [18, Theorem 6], the cumulative distribution function of $\mathbb{P}(S_i^q \leq x|\tau)$ is equal to 0 for $0 \leq x < x_1$, and

$$\hat{h}_n \triangleq \sum_{\substack{\hat{\mathcal{A}} \subseteq \{x_1, \dots, x_n\} \\ \hat{\mathcal{A}} \neq \emptyset}} (-1)^{|\hat{\mathcal{A}}|+1} \prod_{\mathbf{s}_h \in \mathcal{C} \setminus \mathbf{s}_g} \left(1 - \frac{|\hat{\mathcal{A}}|K - e(\hat{\mathcal{A}}, \mathcal{I}_{\mathbf{s}_h})}{L} \right) \quad (31)$$

for $x_n \leq x < x_{n+1}$, $n = 1, \dots, K$, where $x_{K+1} \triangleq \infty$ by convention. Note that the value $e(\hat{\mathcal{A}}, \mathcal{I}_{\mathbf{s}_h})$ can be computed from Lemma 3. More precisely, by letting $\mathcal{A} = \{a - \tau \bmod L : a \in \hat{\mathcal{A}}\}$, we have $e(\hat{\mathcal{A}}, \mathcal{I}_{\mathbf{s}_h}) = e(\mathcal{A}, \mathcal{I}_{\mathbf{s}_h})$, where $\mathcal{A} \subseteq \mathcal{I}_{\mathbf{s}_g}$. Then, the conditional expected value of S_i^q can be computed by

$$\begin{aligned} \mathbb{E}[S_i^q|\tau] &= \sum_{t=0}^{L-1} (1 - \mathbb{P}(S_i^q \leq t|\tau)) \\ &= x_1 + \sum_{n=1}^{K-1} (1 - \hat{h}_n) (x_{n+1} - x_n). \end{aligned} \quad (32)$$

Summing (32) over all $\tau \in \mathbb{Z}_L$, we get

$$\begin{aligned} \mathbb{E}[S_i^q] &= \frac{1}{L} \sum_{\tau=0}^{L-1} \mathbb{E}[S_i^q|\tau] \\ &= \frac{1}{L} \left(\frac{1}{2} C_\delta(0) + \sum_{n=1}^{K-1} (1 - \hat{h}_n) C_\delta(n) \right) - \frac{1}{2}, \end{aligned} \quad (33)$$

where the last identity can be obtained by adapting the proof of [17, Theorem 1].

V. SIMULATION RESULTS

In this section, we evaluate the AoI performance of 2-CRT sequences via numerical simulations over a collision channel with MPR capability 2, with the number of users K ranging from 4 to 15. Each data point is averaged over 10^6 independent runs with random relative delay offsets $\tau \in \mathbb{Z}_L$. We compare the proposed 2-CRT sequences against slotted ALOHA, including a variant with an identical duty factor, i.e., the transmission probability $p_r = K/L = 1/p_K$, to ensure fair resource consumption.

Fig. 1 and Fig. 2 present the AAoI and normalized AAoI, defined by $\bar{A}_i/K$, under generate-at-will and periodic traffic models, respectively. In addition to the simulation data, we include the analytical results calculated using our derived formulas in Sections IV-A and IV-B. The simulation results

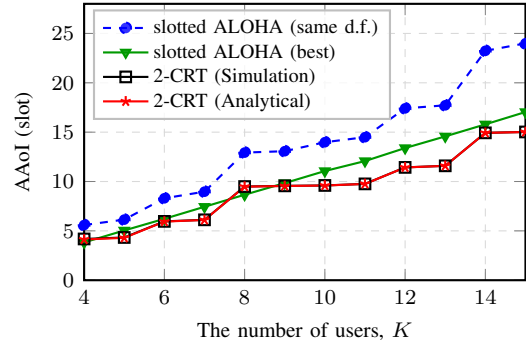


Fig. 1. Average AoI for generate-at-will traffic.

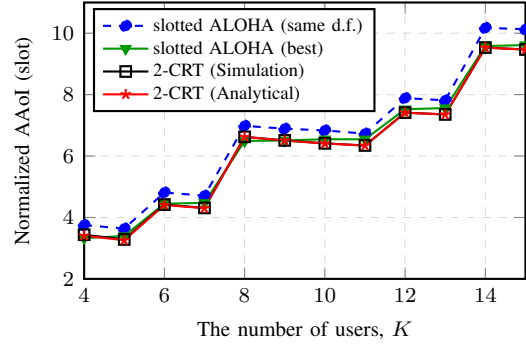


Fig. 2. Normalized average AoI for $T = p_K K$.

are in excellent agreement with the analytical curves, thereby validating the accuracy of our theoretical framework. The results show that 2-CRT sequences consistently outperform all slotted ALOHA variants. This superiority stems from the 2-UI property of 2-CRT sequences.

VI. CONCLUDING REMARKS

This paper investigates the AAoI performance of 2-CRT protocol sequences in a feedback-free collision channel with MPR capability 2. We prove that the proposed sequences possess the 2-UI property, providing a hard guarantee on successful transmissions within a period of time. Through an in-depth analysis of the sequence structure, we derive analytical formulas for AAoI under various traffic models. The simulation results confirm that 2-CRT consistently outperforms slotted ALOHA in terms of AAoI. Future work will consider the general MPR capability $\gamma > 2$. Additionally, we will explore arbitrary update periods T , particularly in cases where T is a factor of L , to investigate how the UI property supports multiple updates within a single period.

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